

Piron Romain | Ph.D

Postdoctoral researcher

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📍 Tokyo, Japan | **Nationality:** French | **Languages:** English (Advanced), French (Native), 日本語 (Beginner)

Profile summary

- Postdoctoral researcher with a background in **theoretical physics and optimization**.
- Experienced in **large-scale simulations of quantum Hamiltonians** and **quantum algorithms**, with hands-on implementation in Python.
- Interested in **applying mathematical modeling to data-driven, real-world systems**.
- Based in Tokyo, with a growing interest in Japanese language and culture

Technical skills

- **Programming and tools:** Python (NumPy, SciPy, Pandas, Matplotlib), MATLAB, Git, Jupyter
- **Analytical:** Complex analysis, Linear Algebra, Quantum systems, Probability, Statistics, Information theory
- **Soft skills:** Communication, Team working, Interdisciplinary collaboration, Rapid learning

Experience

Postdoctoral researcher | Quantum system simulations and algorithmic scalability

📍 国立情報学研究所, Tokyo, Japan 📅 September 2025 - Present

- Develops dimensionality reduction for **scalable simulation frameworks** of high-dimensional quantum interacting models
- Implements **Python pipelines for simulations, data management, and analysis**

Ph.D researcher | Quantum optimization and signal processing

📍 CITI Laboratory, Lyon, France 📅 October 2023 - October 2025

- Designed and implemented **quantum optimization algorithms** for **data inference in large-scale wireless networks**, validated through **Python simulations and experiments on quantum devices**
- Collaborated in an **interdisciplinary research environment** at the intersection of quantum information and signal processing

Event coordinator | Volunteering in student's association

📍 24h de l'INSA, Lyon, France 📅 January 2024 - December 2024

- Led a team of 10 volunteers during **France's largest student-run festival**
- Developed strong **coordination, organization, and leadership skills** in a fast-paced environment

Research internship | Master's thesis in mathematical physics

📍 CERN - European Organization for Nuclear Research, Geneva, Switzerland 📅 February 2023 - July 2023

- Investigated **mathematical consistency conditions** in conformal field theory using the **bootstrap method**, with implications for the AdS/CFT correspondence

Education

Ph.D in Signal processing and optimization

INSA Lyon (Claude Bernard Lyon 1 University), Lyon, France | October 2025

Thesis: Quantum Algorithms for Non-Orthogonal Multiple Access Systems

M.Sc in Physics - Double degree

EPFL (Ecole Polytechnique Fédérale de Lausanne), Lausanne, Switzerland | October 2023

Advanced coursework in quantum field theory, general relativity, and high-energy physics.

M.Sc in Engineering

CentraleSupélec (Paris Saclay University), Gif-sur-Yvette, France | October 2023

Mathematics and physics with interdisciplinary exposure from financial modeling to quantum technologies

Selected publications

(Full list available upon request)

- R.Piron, A.Soeda, "Renormalization Treatment of IR and UV Cutoffs in Waveguide QED and Implications to Numerical Model Simulation, arXiv preprint, 2026
- R.Piron, C.Goursaud, "Scheduling Quantum Annealing for Active User Detection in a NOMA Network" presented at **IEEE QCE, Montréal, Canada, 2024.**